Hormonal therapy for advanced prostate cancer

Hormonal therapies lower testosterone levels or stop testosterone reaching the prostate cancer cells. Hormonal therapy may be given on its own, or in combination with other treatments.

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What is hormonal therapy for advanced prostate cancer?

Hormonal therapy is the main treatment for advanced (metastatic) prostate cancer. It aims to help control the cancer. It can shrink the cancer and reduce symptoms.

Your cancer doctor or specialist nurse will talk to you about the type of hormonal therapy that is suitable for you. They will explain the different side effects, and things that can help manage them.

Prostate cancer needs the hormone testosterone to grow. This is mainly made by the testicles. Small amounts are made by the adrenal glands, above each kidney, and in the body fat. Some hormonal therapy drugs for prostate cancer reduce the amount of testosterone naturally produced in the body. Others stop testosterone from reaching the prostate cancer cells.

Testosterone is important for:

- sex drive (libido)
- getting an erection
- facial and body hair growth
- muscle development and bone strength
- energy
- heart health.

Hormonal therapy to reduce testosterone affects some of these things.

Your cancer doctor or specialist nurse will check how well your hormonal therapy is working. They will ask about your symptoms, examine you and monitor your PSA level. Measuring your PSA level is usually a good guide to how well treatment is working. If your PSA level rises, your cancer team may

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talk to you about having additional treatment.

Surgery to reduce testosterone

Rarely, some people have surgery instead of hormonal therapy, to reduce the amount of testosterone naturally produced by the body.

How hormonal therapy is given

Before you start hormonal therapy, you will have some blood tests. This includes a PSA test. Your doctor will also examine you and ask about your symptoms.

You may have hormonal therapy in the following ways:

- injections
- implants
- a nasal spray
- tablets.

You may have more than 1 type of hormonal therapy. Or you may have hormonal therapy with chemotherapy or radiotherapy when you are first diagnosed with advanced (metastatic) prostate cancer.

These additional treatments can be more effective than hormonal therapy on its own. But there are more side effects to cope with. Your cancer team can tell you more about this.

Androgen deprivation therapy (ADT)

Hormonal therapy is usually given continuously to treat advanced prostate cancer. This is called androgen deprivation therapy (ADT).

ADT works by reducing the amount of testosterone the body produces naturally.

There are 2 types of ADT:

- luteinising hormone-releasing hormone (LHRH) agonists
- gonadotrophion-releasing hormone (GnRH) antagonists.

They work by blocking the message from the brain that tells your testicles to make testosterone.

If you are having ADT as an injection or implant, the nurse at your GP practice can give it to you. Your cancer doctor or specialist nurse will tell them when to start treatment.

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Hormonal therapy injections and implants

LHRH agonists

LHRH agonists are the most common type of ADT. You usually have them as an implant or an injection under the skin (subcutaneously).

The drug buserelin is given as subcutaneous injections for 1 week, and then as daily doses of nasal spray.

The commonly used LHRH agonists are:

- buserelin (Suprefact®) – taken daily
- goserelin (Zoladex®, Zoladex LA®) – given as an injection every 4 or 12 weeks
- leuprorelin (Prostap®, Staladex®) – given as an injection every 4 or 12 weeks
- triptorelin (Decapeptyl®, Gonapeptyl®) – given as an injection every 4 weeks, 3 months or 6 months.

Tumour flare

The first time you have an LHRH agonist, your testosterone levels may go up for a very short time. This could make any symptoms caused by the prostate cancer temporarily worse. This is sometimes called tumour flare. To help prevent tumour flare, your cancer doctor will give you anti-androgen tablets to take for a short time before and after starting the LHRH agonist. Your doctor, nurse or pharmacist will explain more about this.

GnRH antagonists

GnRH antagonists work quicker than LHRH agonists and do not cause tumour flare.

You usually have GnRH antagonists if the cancer has spread to the spine and there is a risk it could press on the spinal cord. This is called metastatic spinal cord compression (MSCC).

You may also have GnRH antagonists if you are at risk of a bone fracture or blockage in the kidney or ureter, or if you have other health conditions.

GnRH antagonists include degarelix (Firmagon®) and relugolix (Orgovyx®). You have degarelix as an injection under the skin of the tummy (subcutaneously) once a month. You take relugolix as tablets.

You may switch from degarelix to an LHRH agonist. This is because you can have LHRH agonists less often than degarelix. Your cancer doctor or specialist nurse can talk to you about this.

Intermittent hormonal therapy

If side effects are difficult to cope with, you may have a break from hormonal therapy before starting it again. This is called intermittent hormonal therapy.

Intermitted hormonal therapy is not suitable for everyone. It may not control the cancer as well as continuous hormonal therapy.

Your cancer doctor or specialist nurse can explain the possible benefits and disadvantages of intermittent hormonal therapy. You will have regular PSA tests during intermittent hormonal therapy. If your PSA level goes up or your symptoms get worse, your cancer doctor or specialist nurse will advise

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starting hormonal therapy again or having continuous treatment with androgen deprivation therapy (ADT).

Hormonal therapy as tablets

Anti-androgen drugs

These drugs stop testosterone from reaching the cancer cells. You take them as tablets. You may have them:

- for 1 to 2 weeks before and after starting an LHRH agonist, to prevent symptoms getting temporarily worse (tumour flare)
- on their own as part of intermittent hormonal therapy
- with other hormonal therapy drugs (ADT) as part of combined hormonal therapy.

Taking an anti-androgen on its own as part of intermittent hormonal therapy may:

- have less impact on your sex drive (libido) and erections than other hormonal therapy drugs
- help avoid the side effects of combined hormonal therapy
- help reduce the risk of long term effects of hormonal therapy, such as bone thinning or increased risk of heart disease.

But anti-androgens alone may not control the cancer as well as other hormonal therapy drugs in some situations.

Anti-androgens include:

- bicalutamide
- cyproterone acetate (Cyprostat®)
- flutamide.

If you have been taking an anti-androgen drug for some months or years, and the cancer begins to grow, your cancer doctor may try stopping the drug. This may make the cancer shrink for a while. Doctors call this a withdrawal response.

Androgen receptor pathway inhibitors (ARPIs)

Androgen receptor pathway inhibitors are given in addition to continuous hormonal therapy (ADT). You take them as tablets. You can take them when you first start ADT or after some time if ADT alone is no longer working for you. You usually take ARPIs for as long as they are working for you and any side effects can be managed.

ARPIs include:

- abiraterone
- enzalutamide (Xtandi®)
- apalutamide (Erleada®)
- ®

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darolutamide (NUBEQA ˜).

Your cancer team can give you more information about these drugs if they are suitable for you.

Abiraterone

You take abiraterone once a day. Side effects include:

- high blood pressure
- build-up of fluid (oedema)
- tiredness (fatigue)
- low level of potassium in the blood
- changes in liver blood tests.

To reduce the risk of side effects, abiraterone is given with a daily dose of a steroid drug called prednisolone. Prednisolone can cause tummy pain or indigestion so you may also get a tablet to help prevent that.

You need to take abiraterone without food. You should not eat for 2 hours before taking abiraterone and for 1 hour afterwards.

You usually take prednisolone with or after food. This means you do not take abiraterone and prednisolone at the same time. It is important to take abiraterone and prednisolone exactly as explained. This is so they work as well as possible for you.

Enzalutamide, apalutamide and darolutamide

If you take 1 of these drugs, you do not need to take prednisolone. You take them daily as tablets.

Side effects include:

- tiredness (fatigue)
- high blood pressure
- risk of falls and fractures.

Less commonly they can affect your memory and concentration. This is called brain fog. Darolutamide is less likely to affect the brain, but it can still happen.

Enzalutamide can affect the liver. This will be checked with regular blood tests. Rarely, it can affect the brain and cause seizures.

Apalutamide can cause a rash. It can also cause an underactive thyroid. This is called hypothyroidism. Symptoms of hypothyroidism include:

- feeling tired
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feeling cold

- gaining weight
- constipation
- low mood.

During apalutamide treatment, you will have regular blood tests to check your thyroid.

Steroids

Sometimes a steroid drug, such as dexamethasone, is used to treat advanced prostate cancer. You usually have it when ADT is no longer working for you.

Testosterone is mainly made by the testicles. A small amount is made by the adrenal glands, above each kidney. Steroids work on the adrenal glands, reducing the amount of testosterone the body produces naturally.

You may have dexamethasone in addition to hormonal therapy (ADT). Side effects of steroids include:

- increased appetite
- weight gain
- difficulty sleeping
- irritability
- thin, fragile skin that bruises easily or takes longer to heal
- bone thinning (osteoporosis).

Oestrogen

Less commonly, treatment with the hormone oestrogen can help to reduce the amount of testosterone the body produces naturally. It may sometimes be used when other types of hormonal therapy are no longer working.

Oestrogen can cause side effects similar to other hormonal therapy drugs for prostate cancer. Oestrogen can also increase the risk of getting a blood clot, so it may not be suitable for some people. To help prevent blood clots, your doctor may prescribe a low dose of aspirin, if suitable for you.

Side effects of hormonal therapy

Different hormonal therapy drugs have different side effects. It is important to get information about the drugs you are having so you know what to expect.

All hormonal therapy drugs affect how the body makes and uses testosterone. The following side effects are common to most hormonal therapy drugs used for prostate cancer.

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Erection difficulties

Most hormonal therapy drugs cause loss of sexual desire (libido) and erection difficulties. This is called erectile dysfunction or ED.

Androgen deprivation therapy (ADT) usually completely stops erections during treatment. Anti-androgens stop erections in most people.

Hot flushes and sweats

Hot flushes are a common side effect. You might have fewer hot flushes as your body adjusts to treatment. If hot flushes do not improve, talk to your cancer doctor, specialist nurse or pharmacist. Certain drugs can help to improve hot flushes. Physical activity can also help.

Anxiety can make hot flushes feel worse. Relaxation techniques might help when you can feel a flush starting.

Tiredness (fatigue)

Feeling tired is a common side effect. It can be made worse by hot flushes affecting your sleep. Try to pace yourself and plan your day so you have time to rest between activities.

Being physically active can help to manage tiredness and give you more energy. It can also help with treatment side effects. Your cancer team can advise you about what physical activity might be suitable.

If tiredness is making you feel sleepy, do not drive or operate machinery.

Breast swelling or tenderness

Over a longer period of time, anti-androgen treatment such as bicalutamide may cause swelling and tenderness of the breast tissue. This is called gynaecomastia. To prevent this, you may have 1 or more low doses of radiotherapy to the chest before treatment starts. You may have another type of hormonal drug to treat the breast swelling. Your cancer doctor or specialist nurse can give you more advice.

Weight gain and loss of muscle strength

You may gain weight with hormonal therapy. This is usually around the tummy. Try to eat a healthy, balanced diet. Regular physical activity such as short walks and exercises can help keep your weight stable. It also helps to look after your muscles. Ask your cancer team for advice about how

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much and how often you should exercise and what activities are suitable for you.

Bone thinning (osteoporosis)

Hormonal therapy can cause bone thinning. This is called osteoporosis. It can sometimes lead to tiny cracks in the bone (fractures). The risk increases if you are taking hormonal therapy for long periods of time. If you are starting long term hormonal treatment, your doctors may arrange for you to have a DEXA scan. This is a dual-energy x-ray absorptiometry scan. The scan allows them to check your bones for any areas of weakness.

To help prevent or reduce bone thinning, you may be given calcium and vitamin D tablets. Your doctor may also prescribe bone-strengthening drugs called bisphosphonates. For example, you may have a drip (infusion) of zoledronic acid once or twice a year, or a weekly tablet called alendronic acid.

Weight-bearing exercises, such as walking, Pilates and yoga, can help look after your bone health. Eating a healthy balanced diet can also help.

Mood changes

You may feel low or depressed, or have mood swings. This can be a side effect of hormonal therapy, but it can also be because you are coping with advanced cancer. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage low mood or depression. They can also support you or refer you to get help with your emotions.

Booklets and resources

Booklets

01 Jul 2023

Coping with fatigue (tiredness)

Audiobooks

28 Jul 2023

Coping with fatigue audiobook

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About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Prostate cancer: diagnosis and management. NICE guideline [NG131] Published: 09 May 2019 Last updated: 15 December 2021 (accessed October 2024) https://www.nice.org.uk/guidance/ng131

Parker, C. et al. Prostate cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology, Volume 31, Issue 9, 1119–1134 (accessed October 2024) https://www.annalsofoncology.org/article/S0923-7534(20)39898-7/fulltext#articleInformation

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Our cancer information meets the PIF TICK quality mark.

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We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

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- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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